Resolution Geometry: Cornering the Floating Box

Protocolism and Realism, Einstein and Newton, Luck and God

Matěj Rada
MatejRada@email.cz
Licensed under CC BY-NC-ND 4.0

October 2026

Abstract

Resolution Geometry (RG) cannot derive general relativity. The source remainder leaves the Weyl curvature free, so the field equation stays a selection or a measurement. What RG can do is *corner* a claim: prove the exact protocol region in which the claim is decided, the region in which it is observationally idle, and the declarations it needs. This paper corners five sets of claims, each with a computed witness.

(1) **The floating box.** Einstein's equivalence principle is a *resolution window*. The connection is removable at a point, the curvature is not. An orbiting gradiometer resolves the tidal tensor at signal-to-noise $\sim 10^{5\text{–}6}$. Even an ideal box reads only the energy footprint: the tidal part R_{0i0j} is free of γ, while the spatial part is proportional to γ. The box is therefore blind to half of light bending: Einstein 1911 gives $0.876''$, general relativity 1915 gives $1.751''$.

(2) **Protocolism against realism.** Every observable task factors through the observable quotient, so the latent surplus of any realism is idle, though not refuted. But protocolism is not self-founding: it needs a declared task and declared units.

(3) **RG against Einstein.** General relativity wins every empirical test RG can run. RG corners its ontology:
- the field equation is selected, not forced;
- manifold points are gauge (the hole argument);
- interiors and topology are remainders;
- G and M appear only as GM.

Einstein's *local* realism in quantum mechanics loses; his determinism survives only nonlocally.

(4) **RG against Newton.** Absolute velocity lies in the kernel, so Leibniz wins there. Absolute rotation lies in the quotient, so Newton wins the bucket. Newton's single potential is exact for flat structure at any field strength and fails exactly where structure is resolved.

(5) **Luck and God.**
- Finite records cannot separate chance from determinism. The deterministic digits of $\sqrt{2}$ pass the same tests as a coin.
- Bell excludes local determinism, not determinism.
- Luck in the kernel is never forgotten.
- A God-observer faces a dilemma. As a protocol it must be contextual (0 of 512 Peres–Mermin assignments), information-limited (one bit per step of chaotic horizon) and unit-conventional. As a non-protocol it lies outside the quotient, and RG is silent.

There is no overall winner in the metaphysical sense. RG holds the upper hand on *what observation decides*. General relativity holds it on *what the physics is*, within its domain. No one wins *what exists*. Battery: 14 arenas, 7 computed canaries rejected, master seed 20261026.

Contents

1 What it means to corner a claim 2

2 The floating box 2

3 Arena I: protocolism against realism 3

4 Arena II: RG against Einstein 4

5 Arena III: RG against Newton 4

6 Arena IV: luck 5

7 Arena V: God 5

8 Scoreboard 6

9 What would move the corners 7

10 Battery 7

11 Non-claims 8

1 What it means to corner a claim

Definition 1.1 (Cornering)**.** Let C be a claim and $\mathcal{P}$ a declared class of protocols. RG *corners* C when it proves three things:

(i) the subclass of $\mathcal{P}$ in which C is decided, by a LOCK or a refutation;

(ii) the subclass in which C lies in a single observational fibre with its rivals (CORRESPONDENCE);

(iii) the declarations that no protocol supplies but C needs.

A party holds the *upper hand* in an arena when its claim survives (i) with the smaller declaration budget (iii). A claim confined entirely to (ii) is *idle*, which is not the same as *refuted*.

Cornering is weaker than refutation and stronger than doubt. It is the only weapon RG has against general relativity, because the physical path paper proved that no function of the source remainder fixes the gravitational field [1]. Every corner below is backed by a computed witness (Section 10). Imported results are marked.

2 The floating box

Einstein's "happiest thought" is the freely falling box, in which gravity disappears. In RG language the box is a protocol: accelerometers, clocks and test bodies inside a region of size L over a time T. The equivalence principle is the claim that this protocol has gravity in its kernel.

Theorem 2.1 (The equivalence principle is a resolution window; F01)**.** *In Fermi coordinates along a freely falling worldline,* $g = \eta + O(R\,x^2)$*. The connection vanishes on the worldline; the curvature does not. For a central mass the tidal tensor is* $\mathrm{diag}(-2,1,1)\,GM/r^3$*, which is traceless. The box lies in the fibre of the inertial box exactly when*

$$|R|\,L < \delta_a \qquad \textit{and} \qquad |R|\,L\,T^2 < \delta_x,$$

where δ_a and δ_x are the declared differential-acceleration and displacement resolutions. In low Earth orbit $GM/r^3 = 1.28\times10^{-6}\,\mathrm{s}^{-2}$, i.e. 1284 E. Gradiometry at the milli-Eötvös level resolves this at signal-to-noise of order 10^5–10^6; space gradiometry of this kind has been flown [6]. Two test masses 1 m apart drift 12.8 mm in 100 s.

Corner. The equivalence principle is exact only in the infinitesimal sense: the Christoffel symbols are gauge, like RG's bisection part. Any finite box with a fine enough instrument resolves gravity, so the box is a CORRESPONDENCE only below a declared resolution.

Theorem 2.2 (The box reads energy, not structure; F02)**.** *For the linearized metric $g = -(1+2\Phi)dt^2 + (1-2\gamma\Phi)\,dx^2$:*

- *the tidal (electric) part $R_{0i0j} = \partial_i\partial_j\Phi$ is independent of γ;*
- *the spatial part R_{ijkl} is proportional to γ.*

Slow probes, clocks and accelerometers read the energy footprint N. Structure h is reached only by light, or by fast probes suppressed by v^2/c^2, exactly as in the duality theorem [2]. At the solar limb the equivalence principle alone (Einstein 1911 [3]) predicts $0.876''$; with structure (1915) the prediction is $1.751''$.

Verdict 2.3 (Floating box). The box is cornered twice. It sees gravity once the resolution is fine enough. Even a perfect box sees only half of gravity, the energy footprint. Einstein could not have derived general relativity from the box: he needed a selected field equation and light. The infinitesimal equivalence principle survives intact, as a gauge statement.

3 Arena I: protocolism against realism

Theorem 3.1 (Realism is idle beyond the quotient)**.** *By universal factorization (L2), every observable task factors through the observable quotient. Latent structure that is constant on fibres has no observable consequence. The following are of this kind:*

- *points of the manifold (F04);*
- *black-hole interiors for exterior protocols (F05);*
- *empirically equivalent interpretations of quantum mechanics (F12).*

Theorem 3.2 (Protocolism is not self-founding)**.** *Pure operationalism needs declarations that no protocol supplies:*

- *a task, since minimal panels are task-relative and no task-free equivariant selector exists (L13, F13);*
- *units, since there is no internal unit (L94, F07);*
- *an admissible class, since there is no uniform forgetting without one (L53);*
- *the hypotheses that open walls (H-reg, H-SI, and so on).*

Verdict 3.3 (Arena I). Protocolism has the upper hand on knowledge: it computes exactly what can be decided. Realism keeps the metaphysical field, but wherever it exceeds the quotient it is idle, not refuted. Bridgman's operationalism [8] needs RG's declarations to stand. The dispute becomes a ledger, not a war.

4 Arena II: RG against Einstein

Einstein's physics. General relativity wins every empirical test that RG's own gravitation stack runs: Mercury's perihelion, light deflection, the GPS clock and Pound–Rebka. The parameter γ is measured to $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [4], and the equivalence principle has been tested at $\eta \sim 10^{-15}$ [5]. RG has no rival dynamics.

Einstein's ontology: the best undermines.

	Claim of general relativity	RG corner	Witness
U1	The field equation follows from gravitation and matter	selected (Lovelock) or measured; no function of the source remainder fixes curvature	F06, P12
U2	The box (equivalence principle) grounds the theory	the box reads energy only; structure needs light	F02
U3	Spacetime points are physical	points are gauge; point coincidences are the observables (hole argument [7])	F04
U4	The metric describes the interior and the global shape	interiors are exterior correspondences; topology is a global remainder	F05, L119
U5	Matter obeys energy conditions	declared in general relativity; forced only for kinetic sources; Λ, the stiff fluid and electric tension lie outside the kinetic cone	Phys. Path P04
U6	G and M are physical constants of the geometry	geometry knows only GM/c^2; $\kappa = 8\pi G$ is a unit anchor	F07
U7	Distance is a property of spacetime	distance is protocol-relative: radar $\neq$ proper $\neq$ optical	Light L09

Einstein's quantum realism. No noncontextual value table exists: 0 of 512 assignments satisfy the Peres–Mermin square (F09). Local deterministic strategies obey $|\mathrm{CHSH}| \leq 2$ (F10). Einstein's *local* realism therefore loses. His determinism does not: a deterministic, no-signalling but nonlocal model reproduces the singlet exactly, with $|S| = 2.829$ (F10).

Verdict 4.1 (Arena II). Physics: **Einstein**. Ontology beyond the quotient (U1–U7): **RG**, which corners rather than refutes. Local realism: **against Einstein**. Determinism ("God does not play dice"): **undecided**, a CORRESPONDENCE. Einstein's own resolution of the hole argument, by point coincidences, is protocolist; in that sense the ontology corner is *Einstein against Einstein.*

5 Arena III: RG against Newton

Theorem 5.1 (Absolute space split; F03)**.**

- *Uniform velocity is in the kernel. Light clocks along and across the motion both give proper round trip $2L/c$, for every v.*
- *Rotation is in the quotient. The Sagnac delay $4A\Omega/c^2$ (1.0×10^{-20} s for a 1 m ring at the Earth's rate) is detected by ring lasers (L105).*

Theorem 5.2 (Newton's exact failure point; F14, imported from the duality paper)**.** *Newton's single potential is* exact *for flat structure at any field strength, with $\Phi_{\mathrm{kin}} = \frac{c^2}{2}(1 - N^{-2})$. It fails*

exactly where structure is resolved. Structure's share of a deflection is $\gamma v^2/c^2$*, which is* 10^{-8} *for planets and* 1 *for light. Two-speed panels refute Newton iff the relative precision is below* $\varepsilon_{\text{crit}}$*; for a slow probe plus light,* $\varepsilon_{\text{crit}} = \gamma/(2+\gamma) = 1/3$*. Newton therefore fails for light in arbitrarily weak fields, and slow probes protect him.*

Verdict 5.3 (Arena III). Absolute position and velocity: **Leibniz and RG** [9]. The bucket [10], i.e. absolute rotation: **Newton**. Gravity: **Newton** for slow probes and flat structure; **Einstein** for light and wherever structure is resolved. Newton's *hypotheses non fingo* is the most protocolist line in the history of physics.

6 Arena IV: luck

Theorem 6.1 (Finite records do not certify chance; F11)**.** *Take 20 000 bits. The binary digits of* $\sqrt{2}$*, which are fully deterministic, and a seeded coin both pass block-frequency tests for* $k = 1, \ldots, 4$*, with all* $p > 0.01$*. Deterministic chaos turns finite information into algorithmically random records at rate* λ *(Brudno [11]; F08). No uniform forgetting and operator-only locks (L52–L56) mean that a sample can refute a declared class but never certify randomness.*

Theorem 6.2 (Kernel luck is permanent; F11, L49)**.** *Two priors on an observed parameter merge as* $1/n$*: their gap is* 1.0*,* 0.18*,* 0.020 *and* 0.002 *at* $n = 1, 10, 100, 1000$*. On an unobserved parameter they stay* 1.0 *apart at every* n*. The luck that sits in the kernel is never forgotten.*

Theorem 6.3 (Bell does not certify luck; F10)**.** *Bell's theorem [12] excludes local determinism. It does not exclude:*

- *nonlocal determinism (Bohm [13]; the explicit F10 model);*
- *models that violate measurement independence.*

Measurement independence, the "free choice" of settings, is itself a declaration and not an observable. Fine's theorem [14] places hidden values exactly where a joint law exists (L43).

Verdict 6.4 (Arena IV). Luck is cornered into three parts:

(a) a declaration: probability from exchangeability (de Finetti, L50);

(b) a permanent kernel residue;

(c) chaotic amplification of finite information.

Whether nature is *ontologically* random or deterministic is a CORRESPONDENCE: no protocol decides it. Luck neither wins nor loses. It is typed.

7 Arena V: God

Here "God" means the omniscient observer of physics: Newton's sensorium, Laplace's demon [16], and Einstein's God who does not play dice. We make no theological claim.

Theorem 7.1 (The omniscience dilemma)**.** *An omniscient observer either is a protocol or is not.*

(A) ***If a protocol****, it is bound by RG's theorems:*

- *it is* contextual*: no noncontextual value table exists (Kochen–Specker [15]; 0/512, F09);*

- *it is* information-limited*: each certified chaotic step costs $\lambda/\ln 2$ bits, and a demon with B bits sees about B steps of the logistic map (horizons $7, 17, 27, 37, 47$ at $B = 10, \ldots, 50$; F08);*
- *it is* unit-conventional*: geometry knows only GM/c^2, so even complete laws fix no metre (F07);*
- *it is* task-declaring*: no task-free selector exists (L13).*

An infinite-precision demon is an infinite protocol, which is a declaration and not an observation.

(B) ***If not a protocol****, it lies outside every observation quotient. RG can neither find nor refute it.*

Verdict 7.2 (Arena V). God is cornered to the boundary of the quotient. This is as far as an observation geometry can go. Newton's sensorium, absolute space, is in the kernel (Arena III). Einstein's non-dicing God is a CORRESPONDENCE (Arena IV). Laplace's demon is a protocol with an infinite information budget. RG does not decide God; it decides what any God-protocol would have to be.

8 Scoreboard

Arena	Claim	Status	Upper hand	Witness
Box	equivalence principle for a finite box	window	resolution decides	F01
Box	the box grounds general relativity	cornered	RG (energy only)	F02
I	realism beyond the quotient	idle	protocolism (knowledge)	F04, F05, F12
I	protocolism self-founding	refuted	declarations needed	F13, F07
II	physics of gravitation	locked by data	Einstein	[4]
II	field equation forced	no-go	RG	F06
II	points, interiors, topology physical	idle	RG	F04, F05
II	local realism (QM)	refuted	against Einstein	F09, F10
II	determinism	correspondence	none	F10
III	absolute velocity	kernel	Leibniz / RG	F03
III	absolute rotation	quotient	Newton	F03
III	Newtonian gravity	exact failure point	Newton (slow, flat h) / Einstein (light)	F14
IV	chance certified by records	refuted	none (correspondence)	F11
IV	kernel luck forgotten	refuted	permanent residue	F11
V	noncontextual omniscience	refuted	contextual	F09
V	finite demon forecasts forever	refuted	one bit per step	F08
V	God as non-protocol	outside	undecided	—

Is there an overall winner?

- On *what observation decides*, RG holds the upper hand in every arena. This is partly by construction: RG is the arbiter of that question, so the victory is methodological, not physical.
- On *what the physics is*, general relativity beats Newton wherever structure is resolved, and RG concedes, having no rival dynamics.
- On *what exists*, nobody wins by observation. Realism is idle, not refuted; God and ontological luck lie at the boundary of the quotient.

The honest summary: *RG is the referee, not a player, except in the box, where it scores.*

9 What would move the corners

1. An RG-native protocol whose remainder *contains* the Weyl data (tidal or geodesic-deviation panels), together with a proof that its dynamics are forced. This would turn the no-go of Arena II into a derivation. It is open.
2. A functional two-speed deflection panel. This would measure structure and energy separately, as an experimental realization of the duality theorem.
3. Bell tests that declare measurement independence explicitly as a hypothesis, typing the superdeterministic fibre.
4. Gradiometric box experiments reported as resolution windows rather than as tests of a principle.

10 Battery

`rg_corner_attacks.py` (NumPy, SymPy, mpmath; master seed 20261026) runs 14 arenas, F01–F14. The computed witnesses are:

- the tidal tensor and the PPN Riemann split (sympy);
- light-clock invariance and Sagnac;
- the hole-diffeomorphism invariance;
- scaling of Schwarzschild and Kepler;
- chaotic horizons at 80-digit precision;
- exhaustive Peres–Mermin and local CHSH;
- a deterministic nonlocal singlet model by Monte Carlo (4×10^5 samples);
- block-frequency tests and kernel posteriors.

F05, F06 and F14 are imported, from Birkhoff, the physical path and the duality paper; their numeric witnesses are by construction. F12 is definitional. Seven canaries are rejected by computed criteria. They state that:

- the equivalence principle is exact for a finite box;
- the equivalence principle fixes light bending;

- absolute velocity is detectable;
- a demon with finite bits forecasts forever;
- a noncontextual omniscient table exists;
- a Bell violation proves indeterminism;
- finite records certify randomness.

Limits. The battery is not blind. Gradiometer sensitivities are order-of-magnitude, and no experimental data are reanalysed. The cornering is only as strong as the declared protocol classes.

11 Non-claims

1. General relativity is not refuted, and no rival dynamics is proposed.
2. No theological claim is made; "God" denotes the omniscient observer of physics.
3. Realism is not refuted. It is shown idle beyond the quotient.
4. The historical attributions are summaries, not exegesis.

References

[1] M. Rada, *Resolution Geometry: The Physical Path*, 2026.

[2] M. Rada, *Resolution Geometry: The Duality of Gravity*, 2026.

[3] A. Einstein, Über den Einfluß der Schwerkraft auf die Ausbreitung des Lichtes, *Ann. Phys.* 35 (1911) 898.

[4] B. Bertotti, L. Iess, P. Tortora, *Nature* 425 (2003) 374.

[5] P. Touboul et al., MICROSCOPE mission: final results of the test of the equivalence principle, *Phys. Rev. Lett.* 129 (2022) 121102.

[6] R. Rummel, W. Yi, C. Stummer, GOCE gravitational gradiometry, *J. Geodesy* 85 (2011) 777.

[7] J. Earman, J. Norton, What price spacetime substantivalism? The hole story, *Brit. J. Phil. Sci.* 38 (1987) 515.

[8] P. W. Bridgman, *The Logic of Modern Physics*, Macmillan, 1927.

[9] G. W. Leibniz, S. Clarke, *Correspondence* (1715–1716).

[10] I. Newton, *Philosophiæ Naturalis Principia Mathematica*, Scholium to the Definitions, 1687.

[11] A. A. Brudno, Entropy and the complexity of the trajectories of a dynamical system, *Trans. Moscow Math. Soc.* 44 (1983) 127.

[12] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics* 1 (1964) 195.

[13] D. Bohm, A suggested interpretation of the quantum theory in terms of "hidden" variables, *Phys. Rev.* 85 (1952) 166.

[14] A. Fine, Hidden variables, joint probability, and the Bell inequalities, *Phys. Rev. Lett.* 48 (1982) 291.

[15] S. Kochen, E. P. Specker, The problem of hidden variables in quantum mechanics, *J. Math. Mech.* 17 (1967) 59.

[16] P.-S. Laplace, *Essai philosophique sur les probabilités*, 1814.